

Youngmin Jo

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Education

Sogang University, Seoul, South Korea

B.S. Mechanical Engineering, Summa Cum Laude (expected)

3/2024 – 8/2027

Major GPA: 4.26/4.3, Overall GPA: 4.17/4.3

Military Leave of Absence, ROK Army, Chinook Crew Member

3/2022 – 6/2023

Research Experience

Undergraduate Researcher

12/2025 – present

Materials Design and Process Engineering Laboratory, **Sogang University**

Advisor: Prof. Heonjae Jeong

Origin of Negative Photoconductivity in Amorphous ZnON (2/2026 – 8/2026)

- Performed hybrid DFT (PBE0) and AIMD calculations on amorphous ZnON to determine the origin of negative photoconductivity
- Traced negative photoconductivity to two coexisting trap channels: N 2p-derived hole localization and low-coordination Zn-derived electron localization
- Quantified their trap energies at the valence/conduction band edges

Thermodynamic Stability of Doped LLZO Solid-Electrolyte Interfaces (7/2026 – present)

- Evaluating interfacial thermodynamic stability and Li-ion conductivity of Al-, Ga-, Ta-, and Sr+Ta-doped LLZO solid electrolytes at the LiSi anode and FeF₃ cathode interfaces for thermal-battery applications
- Assessing dopant-dependent interfacial phase stability and Li-ion transport under realistic operating conditions (500 °C) using first-principles calculations and a machine learning interatomic potential (MLIP)

Synthesis of Fluoride-Ion Solid Electrolytes Derived from Inverse Design (8/2026 – present)

- Benchmarking the group's first experimental workflow by synthesizing β -PbSnF₄ and validating its ionic conductivity against reference values before attempting inverse-design candidate compositions

In Preparation

Y. Jo, M. Heo, H. Jeong, et al., “Origin of Negative Photoconductivity in Amorphous ZnON” (제목 미정)

Research Interests

First-principles and machine-learning approaches to thermal and charge transport in disordered and heterogeneous materials, with emphasis on point defects and interfaces

Technical Projects

Computational Workflow Automation for VASP

7/2026 – 9/2026

- Built a GPU utilization monitor for a shared external compute server, reporting per-GPU load and per-user occupancy
- Filled a monitoring gap left by the HPC scheduler, giving the group a way to check load before submitting jobs
- Added automated logging of calculation parameters, outputs, and completion/failure notifications across three compute servers

Skills: Python, Linux/Bash, REST APIs

Area-Selective ALD: A Computational Screening Approach

3/2026 – 6/2026

Manufacturing Processes

- Surveyed the limitations of conventional deposition methods and examined Area-Selective ALD as an emerging solution, along with its deposition mechanism
- Introduced first-principles (VASP) high-throughput screening as a computational tool that evaluates precursor–substrate interactions prior to experimental deposition, reducing cost while increasing throughput
- Illustrated the approach with a toy model comparing reaction energies of two precursor–substrate systems, distinguishing an energetically unfavorable case from one in which deposition stabilizes the system

Skills: VASP, DFT, ALD process analysis

Automated Robotic Arm Teleoperation System

3/2026 – 6/2026

- Developed a motion-driven robotic arm system capable of real-time teleoperation, aimed at automating repetitive lab experiments and manual labor tasks
- Streamlined the motor control and teleoperation pipeline by integrating the LeRobot package, replacing traditional hard-coding methods

Skills: LeRobot package, Computer vision, Linux, Autodesk Inventor, 3D printing

Structural and Thermal Evaluation of Composite Fishing Rod Materials

9/2025 – 12/2025

Fundamentals of Engineering Materials (course project)

- Investigated whether combining two composite materials could improve the elastic and structural performance of a fishing rod, using ANSYS simulation
- Performed an impact analysis by varying the velocity of a steel ball striking the rod structure to determine the load threshold at failure, showing an improvement over a conventional single-material design
- Conducted a supplementary thermal analysis to evaluate the thermal expansion coefficients of the composite materials across the temperature range relevant to marine field conditions

Skills: ANSYS (structural & thermal analysis), composite materials

Predictive Weather Navigation System

8/2025 – 12/2025

C Programming Basic

- Developed a dynamic route-planning application in C that predicts weather conditions at specific waypoints along a route
- Synchronized spatial coordinates from a navigation API with temporal data from a meteorological API
- Engineered a data processing pipeline to extract, parse, and integrate raw external datasets into a functional format

Skills: C, REST APIs (Navigation & Weather)

Design and Fabrication of a Linear-Actuator Camera Tracking Slider

3/2025 – 6/2025

Machine Production Practice

- Designed a linear-rail camera slider frame using 3D CAD, built from an aluminum block and bolted joints
- Selected and mounted a linear actuator motor to drive the camera along the rail
- Implemented Arduino-based motor control to regulate actuator speed, direction, and reciprocating motion for smooth camera tracking shots

Skills: 3D CAD, Arduino

Awards & Scholarships

President's Award & Dean's List with Highest Honors (Top 1%), Sogang University	Spring 2026
President's Award & Dean's List with Highest Honors (Top 1%), Sogang University	Fall 2025
Encouragement Prize, 11th DB GAPS Investment Competition, DB Financial Investment	10/2025
Best Team Award, DS Division, Samsung Electronics, South Korea	8/2025
Ilju Academy and Culture Foundation Scholarship, Taekwang Group, South Korea	8/2025 – 12/2027
Alumni Association Scholarship, Sogang University	3/2025

Leadership Experience

President (previously Vice President, Executive Committee Member)

3/2024 – 6/2027

MECHA, Mechanical Engineering Society, Sogang University

Direct member engineering projects and manage a technical corporate collaboration with Taejin Technology (HTC KOREA)

Team Leader, Samsung Shining Star 2025, Samsung Electronics, South Korea 8/2025

Led a project team through a training program focused on semiconductor foundry processes and the eight major fabrication steps, and presented the team's final project

Teaching & Outreach

English Tutor (Volunteer) | Local Children's Center, South Korea 9/2025 – 6/2027

Deliver biweekly English lesson plans and academic mentorship to five middle school students

Technical Seminar Instructor | MECHA, Sogang University 3/2026

Designed and delivered a seminar on Arduino-based hardware prototyping

Technical Seminar Instructor | MECHA, Sogang University 5/2025

Designed and delivered a seminar on 3D modeling and printing

Global Volunteer (Exchange Participant) | AIESEC in FHCMC, Vietnam 12/2024 – 2/2025

Taught English to secondary students and facilitated Korean language and cultural exchange sessions for university students over 210 hours of service.

Technical Skills

Computational Materials Design: VASP, Gaussian, Pymatgen, ASE, Avogadro

Programming & Systems: Python, C, Linux/Bash, REST APIs

Engineering & Design: Arduino, MATLAB, AutoCAD, Autodesk Inventor, 3D Printing